

Appendix D: Linear Motion

Appendix D: Linear Motion System Specifications

D.1 Ball Screw Specifications

D.1.1 Ball Screw Accuracy Grades (ISO 3408-3, DIN 69051)

Grade	Lead Accuracy (mum/300mm)	Applications	Cost
C10 (Precision 10)	+/-52	General industrial, low precision	Low cost
C7 (Precision 7)	+/-52	Standard CNC, general machining	Medium cost
C5 (Precision 5)	+/-23	Precision CNC, coordinate measuring	High cost
C3 (Precision 3)	+/-12	High-precision CNC, grinding machines	Very high cost
C2 (Precision 2)	+/-8	Ultra-precision machining, inspection equipment	Very high cost
C1 (Precision 1)	+/-4	Semiconductor equipment, metrology	Very high cost

Nomenclature: Lead error = maximum deviation over 300mm travel length

Example: C5 grade ball screw with 2m travel -> max cumulative error = $2000/300 \times 23 = 153$ mum (~0.15mm)

Recommendation for CNC Gantry: - **C7:** Plasma, flame cutting, router (+/-0.1mm tolerance acceptable) - **C5:** Milling, waterjet (+/-0.05mm tolerance required) - **C3:** Precision milling, engraving, PCB drilling

D.1.2 Ball Screw Sizing and Nomenclature

Standard Designation: RM2005-C7 - **R:** Rolled ball screw (ground = “G”) - **M:** Metric - **20:** Nominal diameter (mm) of screw shaft - **05:** Lead (mm) per revolution - **C7:** Accuracy grade

Common Diameters and Leads:

Designation	Diameter (mm)	Lead (mm)	Critical Speed (RPM) @ 1m	Max Velocity (mm/s) @ 3000 RPM	Typical Use
RM1204	12	4	4500	200	Light duty, Z-axis
RM1605	16	5	3800	250	Small CNC, 3D printer Z
RM2005	20	5	3200	250	Standard CNC X/Y
RM2010	20	10	3200	500	Fast rapids, plasma X/Y
RM2505	25	5	2800	250	Heavy CNC X/Y/Z
RM2510	25	10	2800	500	Heavy CNC, fast rapids
RM3205	32	5	2400	250	Very heavy loads
RM3210	32	10	2400	500	Large gantry X-axis

Critical Speed: Speed at which screw begins to resonate/whip (function of diameter, unsupported length, end fixation).

$$\omega_{crit} = \frac{\pi}{L^2} \sqrt{\frac{E \cdot I}{m}}$$

where: - L = unsupported length (m) - E = Young's modulus of steel (210 GPa) - I = moment of inertia of screw shaft (m^4) - m = mass per unit length (kg/m)

Safety Factor: Operate below 80% of critical speed to avoid vibration.

D.1.3 Ball Screw Load Ratings and Life

Dynamic Load Rating (C_a): Load (in N) at which screw achieves 1 million revolutions (L_{10} life).

Life Calculation:

$$L_{10} = \left(\frac{C_a}{F_a} \right)^3 \text{ (million revolutions)}$$

where: - L_{10} = rated life (90% of screws survive) - C_a = dynamic load rating (N) - F_a = applied axial load (N)

Convert to hours of operation:

$$L_{hours} = \frac{L_{10} \times 10^6}{RPM \times 60}$$

Example: RM2010 ball screw - C_a = 8,500 N (typical for RM2010) - Applied load: F_a = 1,000 N (gantry + cutting force) - Operating speed: 1500 RPM average

$$L_{10} = \left(\frac{8500}{1000} \right)^3 = 614 \text{ million revolutions}$$

$$L_{hours} = \frac{614 \times 10^6}{1500 \times 60} = 6,822 \text{ hours}$$

At 40 hours/week operation -> 170 weeks (3.3 years) before replacement

D.1.4 Ball Screw Preload Methods

Purpose of Preload: Eliminate axial play (backlash), increase stiffness.

Preload Methods:

Method	Description	Stiffness	Backlash	Cost
Single Nut (No Preload)	Standard nut with internal clearance	Low	0.05-0.15mm	Low cost
Double Nut (Spacer Preload)	Two nuts with shim spacer	Medium	<0.01mm	Medium cost
Double Nut (Spring Preload)	Two nuts with compression spring	Medium	<0.01mm, adjustable	Medium cost
Oversized Ball Preload	Single nut with larger balls	High	<0.005mm	High cost
Gothic Arch Preload (Ground)	Special ball groove geometry	Very High	<0.002mm	Very high cost

Recommended Preload: - **CNC milling/routing:** 3-8% of dynamic load rating (medium preload)
- **Plasma/waterjet:** 1-3% of dynamic load rating (light preload, lower friction) - **Grinding/ultra-precision:** 8-12% of dynamic load rating (heavy preload)

Formula:

$$F_{preload} = 0.03 \times C_a \text{ to } 0.08 \times C_a$$

D.1.5 Ball Screw End Fixation

Fixation Type	Axial Stiffness	Critical Speed	Cost	Applications
Free-Free	Very Low	Low	Low cost	Not recommended (only for prototyping)
Fixed-Free	Low	Medium	Low cost	Short screws (<500mm), low loads
Fixed-Supported	Medium	Medium-High	Medium cost	Standard CNC (<1500mm)
Fixed-Fixed	High	High	High cost	Long screws (>1500mm), high-speed rapids

Fixed End: BK/BF bearing blocks with angular contact bearings (preloaded), constrained axially and radially.

Supported End: FK/FF bearing blocks with deep groove ball bearings, constrained radially only (allows thermal expansion).

Example Setup for 2m X-Axis: - Motor end: BK20 (fixed, locked with locknut) - Opposite end: FK20 (supported, allows 0.5mm axial float) - Thermal expansion @ 20 degC temp rise: $\Delta L = 2000 \times 11 \times 10^{-6} \times 20 = 0.44 \text{ mm}$ -> supported end accommodates without stress

D.2 Linear Guide Rail Specifications

D.2.1 Linear Guide Types

Type	Load Direction	Rigidity	Accuracy	Cost	Applications
Square Rail (THK HRW, HIWIN HGW)	Heavy radial/moment	Highest	+/-5mum	Very high cost	Heavy milling, large gantry
Standard Rail (THK HRC, HIWIN HGH)	Moderate all directions	High	+/-10mum	High cost	General CNC, mid-size machines
Compact Rail (THK SRG, HIWIN MGN)	Light to medium	Medium	+/-15mum	Medium cost	Small CNC, 3D printers
Miniature Rail (THK SSR, HIWIN EGH)	Light radial only	Low	+/-20mum	Low cost	Pick-place, linear actuators
Cylindrical Rail (SBR)	Light radial	Low	+/-50mum	Low cost	Budget CNC, not recommended for precision

Accuracy Grades: - **Normal (N):** +/-10-15mum straightness/parallelism over 300mm - **High (H):** +/-5-8mum straightness/parallelism over 300mm - **Precision (P):** +/-3-5mum straightness/parallelism over 300mm - **Super Precision (SP):** +/-2mum straightness/parallelism over 300mm

D.2.2 Linear Guide Load Ratings

Static Load Rating (C_0): Maximum static load before permanent deformation (safety factor 3-5 typical).

Dynamic Load Rating (C): Load at which guide achieves 50 km of travel (L_{50} life).

Life Calculation:

$$L = \left(\frac{f_H f_T f_C f_W C}{P} \right)^3 \times 50 \text{ km}$$

where: - f_H = hardness factor (1.0 for standard) - f_T = temperature factor (1.0 for <100 degC) - f_C = contact factor (1.0 for recirculating ball) - f_W = load factor (1.0-1.5 depending on direction) - C = dynamic load rating (N) - P = applied load (N)

Example: HIWIN HGW25CC (Heavy Square Rail) - Dynamic load rating: $C = 60,800$ N - Applied load (vertical axis with gantry): $P = 5,000$ N - Load factors: All = 1.0 (conservative)

$$L = \left(\frac{60800}{5000} \right)^3 \times 50 = 9,083 \text{ km}$$

At 200 mm/s average speed:

$$L_{\text{hours}} = \frac{9,083,000}{0.2 \times 3600} = 12,615 \text{ hours (6.3 years @ 40 hrs/week)}$$

D.2.3 Common Linear Guide Sizes and Ratings

Standard Rail (HG-Series, equivalent to THK HRC):

Rail Size	Block Type	Dynamic Load (N)	Static Load (N)	Preload Options	Weight/Block (kg)	Typical Axis
HGH15CA	15mm compact	12,240	17,160	Z0, ZA, ZB	0.14	Small CNC Z-axis
HGH20CA	20mm compact	24,990	37,730	Z0, ZA, ZB	0.29	Mid CNC Y/Z-axis
HGH25CA	25mm standard	38,220	59,290	Z0, ZA, ZB	0.66	Standard CNC X/Y
HGH30CA	30mm standard	54,930	88,200	Z0, ZA, ZB	1.08	Heavy CNC X/Y
HGW35CC	35mm heavy	94,080	191,820	ZA, ZB, ZC	2.15	Large gantry X-axis
HGW45CC	45mm heavy	156,960	343,350	ZA, ZB, ZC	4.22	Very heavy milling

Preload Classes: - **Z0:** No preload (light clearance), smooth motion, higher speed - **ZA:** Light preload, low friction, 3-6% of C_0 - **ZB:** Medium preload, standard CNC, 6-10% of C_0 - **ZC:** Heavy preload, high rigidity, 10-13% of C_0

Recommendation: ZA for plasma/waterjet, ZB for milling/routing.

D.2.4 Linear Guide Mounting and Alignment

Mounting Bolt Torque (Grade 12.9):

Rail Size	Bolt Size	Torque (N*m)	Bolt Pattern Spacing (mm)
15mm	M4	2.5	60
20mm	M5	4.5	60
25mm	M6	7.5	60

Rail Size	Bolt Size	Torque (N*m)	Bolt Pattern Spacing (mm)
30mm	M8	18	80
35mm	M10	36	80
45mm	M12	62	100

Alignment Procedure: 1. Surface grind or precision machine mounting surface (flatness <0.02mm per 300mm) 2. Clean surfaces with acetone (remove all oil/debris) 3. Apply thin oil film to rail bottom 4. Tighten mounting bolts to 50% torque in sequence from center outward 5. Install blocks, check for smooth motion (no tight spots) 6. Torque bolts to 100% spec 7. Verify running parallelism: <0.02mm over full travel (dial indicator on reference surface)

Parallelism Between Rails (Dual Rail Axis): - **Y-axis (gantry beam):** <0.02mm over full length (critical to prevent binding) - **X-axis (dual rails on same beam):** <0.01mm over full length

D.3 Rack and Pinion Systems

D.3.1 Rack Types and Specifications

Type	Pressure Angle	Accuracy	Backlash	Cost	Applications
Standard Spur Gear Rack	20 deg	+/-0.1mm/m	0.1-0.3mm	Low cost	Budget CNC, large travel low precision
Ground Spur Rack	20 deg	+/-0.05mm/m	0.05-0.1mm	Medium cost	Standard CNC, long-travel plasma
Helical Rack	19.5 deg-20 deg helix	+/-0.03mm/m	0.02-0.05mm	High cost	High-speed CNC, smooth motion
Ball Bearing Rack	N/A	+/-0.02mm/m	<0.01mm	Very high cost	Precision long-travel (rare, specialty)

Common Modules (Metric): - **M1.5 (1.5mm pitch):** Small CNC, light loads - **M2.0 (2.0mm pitch):** Mid-size CNC - **M2.5 (2.5mm pitch):** Standard plasma tables - **M3.0 (3.0mm pitch):** Heavy routers, large gantry - **M4.0 (4.0mm pitch):** Industrial large-format

Module Calculation:

$$\text{Module (M)} = \frac{\text{Pitch diameter (mm)}}{\text{Number of teeth}}$$

Linear travel per pinion revolution:

$$L = M \times \pi \times N$$

where N = number of pinion teeth.

Example: M2.5 module, 20-tooth pinion

$$L = 2.5 \times \pi \times 20 = 157 \text{ mm/rev}$$

D.3.2 Pinion Selection

Number of Teeth: 16-25 teeth typical (trade-off between size and smoothness)

Recommendations: - **16 teeth:** Minimum (small diameter, higher backlash) - **20 teeth:** Standard (good balance) - **25 teeth:** Smoother motion, larger diameter, lower torque ripple

Backlash: 0.05-0.15mm typical for standard gears. Reduced via: - Split pinion (two pinions with spring preload) - Anti-backlash pinion (internal spring mechanism) - Dual rack (two racks at different heights with opposing pinions)

D.3.3 Rack and Pinion Mounting

Rack Mounting: - Mount to precision-ground steel channel or extrusion - Dowel pins every 500mm for alignment (prevents creep) - Torque mounting bolts to specified value (M6 -> 7.5 N*m typical) - Shim as needed to align pitch line parallel to guide rails (+/-0.05mm over 1m)

Pinion Alignment: - Pinion-to-rack clearance: 0.1-0.2mm (use feeler gauge) - Check mesh across full travel (no tight/loose spots)

D.4 Couplings and Shaft Connections

D.4.1 Flexible Coupling Types

Type	Misalignment (angular/parallel)	Torsional Stiffness	Backlash	Cost	Applications
Oldham Coupling	1 deg / 0.5mm	Medium	<0.1 deg	Low cost	General motion, moderate precision
Bellows Coupling	1 deg / 0.1mm	High	<0.05 deg	Medium cost	Servo/stepper to ball screw (standard)
Beam Coupling	3 deg / 0.25mm	Very High	<0.01 deg	Medium cost	Encoder coupling, high precision
Jaw Coupling (Spider)	1 deg / 0.25mm	Low	0.2 deg	Low cost	Light duty, absorbs shock, vibration
Disc Coupling	1 deg / 0.1mm	Very High	<0.01 deg	High cost	High-speed, high-torque, precision

Recommendation for CNC: - **Bellows coupling** for motor-to-ball screw (most common, zero backlash, accommodates slight misalignment) - **Beam coupling** for encoder mounting (torsionally stiff, electrically isolating)

D.4.2 Coupling Sizing

Torque Rating: Select coupling with torque rating $\geq 2 \times$ motor peak torque (safety factor for shock loads, acceleration).

Bore Sizes: Must match motor shaft and ball screw diameter. Common: - Motor shaft: 8mm, 10mm, 12.7mm (1/2"), 14mm - Ball screw: 10mm, 12mm, 14mm, 16mm

Clamping Method: - **Set screw:** Simple, low cost, can slip under high torque - **Clamp style:** Higher torque capacity, no shaft damage, requires flat on shaft

Example: NEMA 23 (14mm shaft) to RM2010 ball screw (16mm diameter) - Coupling: Bellows type, 8mm and 10mm bores, clamp style - Torque rating: 10 Nm (*NEMA 23 peak torque = 4 Nm* -> 2.5x safety factor)

D.5 Lubrication Specifications

D.5.1 Ball Screw Lubrication

Lubricant Types: - **Lithium grease (NLGI Grade 2):** General purpose, standard CNC - **Molybdenum disulfide (MoS₂) grease:** High load, slow speed - **PTFE grease:** Clean environments (medical, food), low lubricity - **Oil mist system:** High-speed applications (>3000 RPM), automatic

Lubrication Schedule: - **Light duty (plasma, router):** Re-grease every 500 hours - **Medium duty (milling):** Re-grease every 200 hours - **Heavy duty (high load, high speed):** Re-grease every 100 hours or oil mist system

Re-greasing Procedure: 1. Clean old grease from nut exterior 2. Inject new grease via grease nipple (Zerk fitting) 3. Cycle axis back/forth 10-20 times to distribute grease 4. Wipe excess grease from screw shaft 5. Verify smooth motion (no binding)

D.5.2 Linear Guide Lubrication

Automatic Lubrication (Recommended): - Single-shot lubricator (refillable cartridge, activates each block pass) - Centralized lube system (pneumatic pump, distributes to all blocks)

Manual Lubrication: - Oil: ISO VG 32 hydraulic oil (standard) - Apply 2-3 drops per block via grease nipple - Frequency: Every 100km of travel or 6 months, whichever first

Cleaning: Wipe rails weekly with lint-free cloth + light oil. Remove chips/debris that can score rail/ball raceways.

End of Linear Motion System Specifications Appendix